Research and caveats summary

DEMO-EH-1003 · *Khaya senegalensis* · bark · status: Evidence review · **SYNTHETIC DEMO DATA**

Activity: bark collection from farm-boundary trees · Region: Borgou department, Benin (synthetic) · Jurisdiction: Benin · Exact site: —

A Harvest Passport is an internal evidence and traceability record. It is not a government permit, a FairWild certificate, a validated local harvest quota, or an assurance of legal compliance or product safety.

Species- and part-specific cautions

Status: ok

Published observational comparison (harvest not randomised) on bark + foliage harvest (supported-by-study; applies: same species, same part; review: awaiting expert review)

Mean long-term growth rate λ 0.982 in harvested populations vs 1.008 in the comparison populations (difference -0.026; 6 vs 6 matrices).

Next: Use it as a scenario in the Studio; local stage counts and recovery measurements are still needed before setting any limit.

Sources: GAOUE OG, TICKTIN T (2010) Effects of Harvest of Nontimber Forest Products and Ecological Differences between Sites on the Demography of African Mahogany. Conservation Biology 24:605-614. doi:10.1111/j.1523-1739.2009.01345.x (https://doi.org/10.1111/j.1523-1739.2009.01345.x)

Published observational comparison (harvest not randomised) on bark + foliage harvest (supported-by-study; applies: same species, same part; review: awaiting expert review)

Mean long-term growth rate λ 0.975 in harvested populations vs 0.992 in the comparison populations (difference -0.017; 6 vs 6 matrices).

Next: Use it as a scenario in the Studio; local stage counts and recovery measurements are still needed before setting any limit.

Sources: Gaoue OG, Horvitz CC, Ticktin T, Steiner UK, Tuljapurkar S (2013) Defoliation and bark harvesting affect life■history traits of a tropical tree. Journal of Ecology 101:1563-1571. doi:10.1111/1365-2745.12140 (https://doi.org/10.1111/1365-2745.12140)

How sensitive this population is to the harvested process (survival and growth of established plants) (computed-from-matrices; applies: this species; review: shown with caveats; awaiting expert review)

About 99% (95% range 99–100%) of this plant's long-run population growth depends on survival and growth of established plants, the process a bark harvest affects (71 published matrices, this species).

Next: Use this to judge which local measurements matter most (e.g. survival of large plants), not as a harvest limit.

Sources: COMPADRE v6.26.3.0 matrices (computed here)

With good and bad years mixed in (computed-from-matrices; applies: this species; review: computed; awaiting expert review)

Simulating 20,000 years that randomly draw from the published matrices, the harvested populations shrinks (λs 0.9814) and the unharvested ones shrinks (λs 0.9902).

Next: Compare this with local monitoring before drawing conclusions.

Sources: COMPADRE v6.26.3.0 matrices (computed here)

1 paper on this species reports that harvest reduced population size or growth. (ai-found-literature; applies: this species; review: needs expert review before use)

Found by the evidence agent: JEV classified the paper, Groq extracted the finding, and the quote below was checked word-for-word against the abstract.

Next: A scientific reviewer should read the paper(s) and confirm or reject this finding.

Sources: Effects of Harvest of Nontimber Forest Products and Ecological Differences between Sites on the Demography of African Mahogany (https://doi.org/10.1111/j.1523-1739.2009.01345.x)

1 paper on this species reports that harvest increased other outcomes. (ai-found-literature; applies: this species; review: needs expert review before use)

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Sources: Defoliation and bark harvesting affect life■history traits of a tropical tree (https://doi.org/10.1111/1365-2745.12140)

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Sources: Defoliation and bark harvesting affect life■history traits of a tropical tree (https://doi.org/10.1111/1365-2745.12140)

1 paper on other species reports that harvest reduced reproduction (plant part differs or is not stated). (ai-found-literature; applies: other species (transferability uncertain); review: needs expert review before use)

Found by the evidence agent: JEV classified the paper, Groq extracted the finding, and the quote below was checked word-for-word against the abstract.

Next: A scientific reviewer should read the paper(s) and confirm or reject this finding.

Sources: Effects of leaf harvesting intensity on fruiting processes in three fodder tree species in the West African Savannah (https://europepmc.org/article/AGR/IND608893946)

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Sources: Effects of leaf harvesting intensity on fruiting processes in three fodder tree species in the West African Savannah (https://europepmc.org/article/AGR/IND608893946)

1 paper on other species reports that harvest did not change growth. (ai-found-literature; applies: other species (transferability uncertain); review: needs expert review before use)

Found by the evidence agent: JEV classified the paper, Groq extracted the finding, and the quote below was checked word-for-word against the abstract.

Next: A scientific reviewer should read the paper(s) and confirm or reject this finding.

Sources: Functional traits partially mediate the effects of chronic anthropogenic disturbance on the growth of a tropical tree. (https://europepmc.org/article/MED/29991997)

Evidence from another species harvested for bark: Prunus africana (transferred-from-other-species; applies: different species, same part; review: awaiting expert review)

Mean long-term growth rate λ 0.999 in harvested populations vs 0.985 in the comparison populations (difference +0.015; 4 vs 1 matrices).

Next: Look for a study on this species; until then monitor plots before and after harvest.

Sources: Stewart KM (2001) The commercial bark harvest of the African cherry (Prunus africana) on Mount Oku, Cameroon (PhD thesis); matrices as archived in COMPADRE

Evidence from another species harvested for bark: Garcinia lucida (transferred-from-other-species; applies: different species, same part; review: awaiting expert review)

Long-term growth rate by scenario (No bark harvest: λ 1.063; Partial debarking (one side): λ 1.060; Ring-barking (whole circumference): λ 1.057; Felling at 1 m after fruiting: λ 1.057; Felling at 1 m before fruiting: λ 1.056). λ below 1 means a population that shrinks if conditions persist.

Next: Look for a study on this species; until then monitor plots before and after harvest.

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Sources: Guedje NM, Zuidema PA, During H, Foahom B, Lejoly J (2007) Tree bark as a non-timber forest product [Garcinia lucida]. For Ecol Manage 240:1-12 (https://doi.org/10.1016/j.foreco.2006.09.029)

CITES Appendix II listing (regulatory-listing; applies: same species; parts covered depend on the annotation; review: awaiting expert review)

International trade needs CITES documents issued by the exporting country's Management Authority. Annotation: ['[GENUS listing Khaya spp.] African populations.']. The listing does not govern collection itself.

Next: Before any cross-border shipment, confirm with the CITES Management Authority which documents apply to your product.

Sources: Species+ (CITES and CMS listings) (https://speciesplus.net/); CITES Appendices I, II and III (https://cites.org/eng/app/appendices.php)

National collection rules (requires-authority-confirmation; applies: Benin; review: awaiting expert review)

We have not yet located an official source for collection rules that apply to this plant part in Benin.

Next: Ask the national forestry administration which permits apply to bark and foliage collection.

Sources:

6 papers the AI could not classify confidently (needs-review; applies: search results; review: awaiting expert review)

JEV and the keyword rules disagreed, or JEV was unsure. They may or may not contain harvest effects.

Next: A scientific reviewer should screen these and promote relevant findings.

Sources: Traditional medicinal knowledge and diversity of medicinal plants in Benin: a systematic review (2014-2023). (2026) (https://europepmc.org/article/MED/42351099); Traditional Medicinal Plants Used for Cancer Treatment in Sub-Saharan Africa: A Systematic Review. (2026) (https://europepmc.org/article/MED/42357155); Ethnomedicinal knowledge of plants used in alternative medicine as uterotonics in Lubumbashi, Haut-Katanga, Democratic Republic of Congo. (2026) (https://europepmc.org/article/MED/42599903); Impact of Human Activities on Woody Vegetation in Gallery Forests in the Mandara Mountains (Far North, Cameroon). (2024) (https://europepmc.org/article/MED/39444784); Review and cross-cultural analysis of medicinal plants traditionally used in Mauritania. (2025) (https://europepmc.org/article/MED/41437073); How do plant demographic and ecological traits combined with social dynamics and human traits affect woody plant selection for medicinal uses in Benin (West Africa)? (2024) (https://europepmc.org/article/MED/38336725)

Legal compliance is part of sustainable collection (general; applies: any wild collection; review: awaiting expert review)

Voluntary standards treat legal compliance at local, national and international level as a precondition.

Next: List the authorisations your collection needs and attach them to the passport.

Sources: FairWild Standard: Version 3.0 (approved 19 December 2023) (https://www.fairwild.org/s/FairWildStandard-V3-FINAL.pdf)

Customary rights and benefit sharing (general; applies: any wild collection; review: awaiting expert review)

Collection should respect collectors' and communities' customary rights and share benefits fairly.

Next: Record how consent and benefit sharing were agreed with the community that holds rights to the site.

Sources: FairWild Standard: Version 3.0 (approved 19 December 2023) (https://www.fairwild.org/s/FairWildStandard-V3-FINAL.pdf)

Collection and post-harvest quality practice (general; applies: any wild collection; review: awaiting expert review)

GACP guidance covers identification, collection practice, drying, storage and documentation of herbal starting materials.

Next: Keep identification, drying and storage records per lot; ask the buyer which GACP expectations apply.

Sources: WHO guidelines on good agricultural and collection practices (GACP) for medicinal plants (2003) (https://www.who.int/publications/i/item/9241546271); EMA Guideline on Good Agricultural and Collection Practice (GACP) for starting materials of herbal origin (EMEA/HMPC/246816/2005; Revision 1 adopted by HMPC in 2025) (https://www.ema.europa.eu/system/files/documents/scientific-guideline/hmpc-guideline-gacp-may-2025_en.pdf)

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